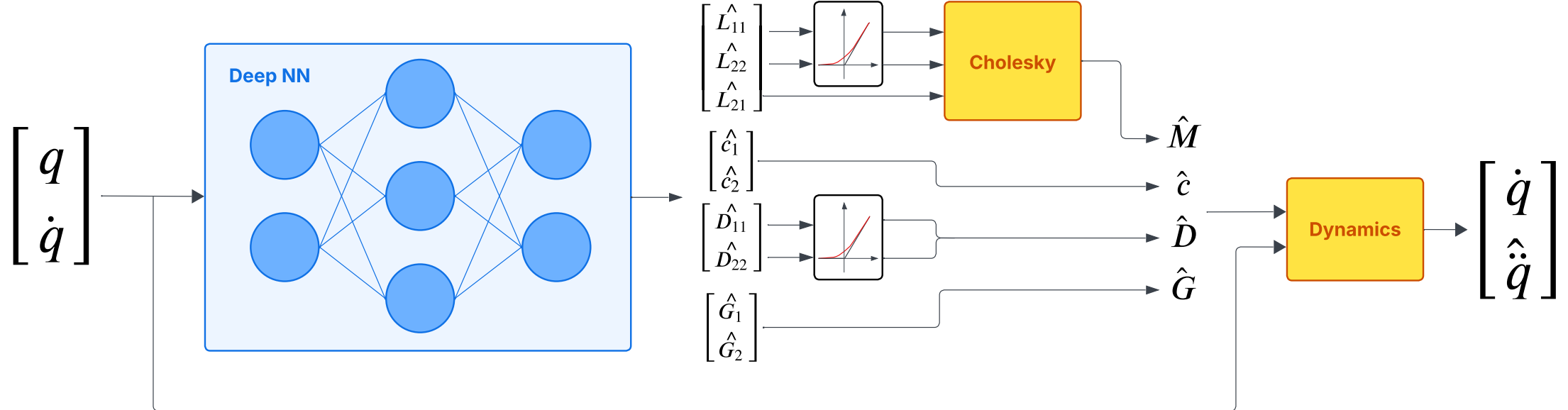
$\begin{bmatrix} q \\ \dot{q} \end{bmatrix}$
Deep NN
$\begin{bmatrix} \hat{L}_{11} \\ \hat{L}_{22} \\ \hat{L}_{21} \end{bmatrix}$
Cholesky
$\hat{M}$
$\begin{bmatrix} \hat{c}_1 \\ \hat{c}_2 \end{bmatrix}$
$\hat{c}$
$\begin{bmatrix} \hat{D}_{11} \\ \hat{D}_{22} \end{bmatrix}$
$\hat{D}$
$\begin{bmatrix} \hat{G}_1 \\ \hat{G}_2 \end{bmatrix}$
$\hat{G}$
Dynamics
$\begin{bmatrix} \dot{q} \\ \ddot{\hat{q}} \end{bmatrix}$